CS4535 · READING R12 · OCT 28, 2026

An Endless Stream of AI Slop

How developers discuss the burden of AI-assisted software development

Baltes, Cheong, Treude · 2026 · arXiv:2603.27249

Writing code got cheap. Checking it didn't.

Naman Rusia · Zhirui

THE HOOK

Our review bot found the paper's example

[R07] · quoted in the paper

“[The agent] hallucinated external services, then mocked out the hallucinated external services.”

A developer describing an AI agent's PR, 2025

PR #936 · Codex review bot

“…requires this nonstandard field while the mock fabricates it in tests/mocks/discord/server.ts, masking the failure.”

Flagged P1 on a 13,577-line AI-assisted PR, Aug 2026

Was the bot right? We checked. Hold that thought.

THE IDEA

Slop: cheap to make, expensive to check

1

Superficial competence

It looks finished. The substance underneath is thin.

2

Asymmetry of effort

It takes far less work to make than to evaluate.

3

Mass producibility

It arrives in volume, because making more costs nothing.

Not the same as a hallucination. Slop can be correct and still waste the reader's time.

Three properties from Kommers et al., “Why Slop Matters” (ACM AI Letters, 2026), which the paper uses as its definition.

THE STUDY

1,154 posts that say “AI slop”

1,154
posts from Reddit and Hacker News

15
threads (12 Reddit, 3 Hacker News result pages)

15
codes, grouped into 3 clusters

234
AI-applied labels a human changed

SEARCH

Exact phrase “ai slop” in three programming subreddits and on Hacker News, Sep 26–27, 2025.

CODE

Open coding on five threads, merged into 15 codes, clustered into three themes.

CHECK

Claude Code applied the labels to every post. A second author checked all of them.

The authors used AI to save time, then had a human check all of it. That's the paper's argument, applied to its own method.

THE CLAIM

A tragedy of the commons



“The development time has been shortened but the team now needs to spend more time to review. Doesn't look like any benefit.”

[R05]

One team in the corpus reported 30 PRs a day across 6 reviewers [R07].

Three clusters of complaints

Review friction

`reviewer-burden · trust-erosion · detection · accountability · mitigations`

- Reviewers get code its author can't explain.
- Feeling like “the first human being to ever lay eyes on this code.”

“It's not AI's code, it's my code.”

Quality degradation

`ai-limitations · codebase-degradation · comprehension-gap · skill-atrophy`

- setTimeout band-aids, casts to any, tests changed to pass.
- An agent that “aborted early in a middleware basically skipping most of AuthZ.”

“You can go very fast with AI, but you accrue technical debt at a much higher speed too.”

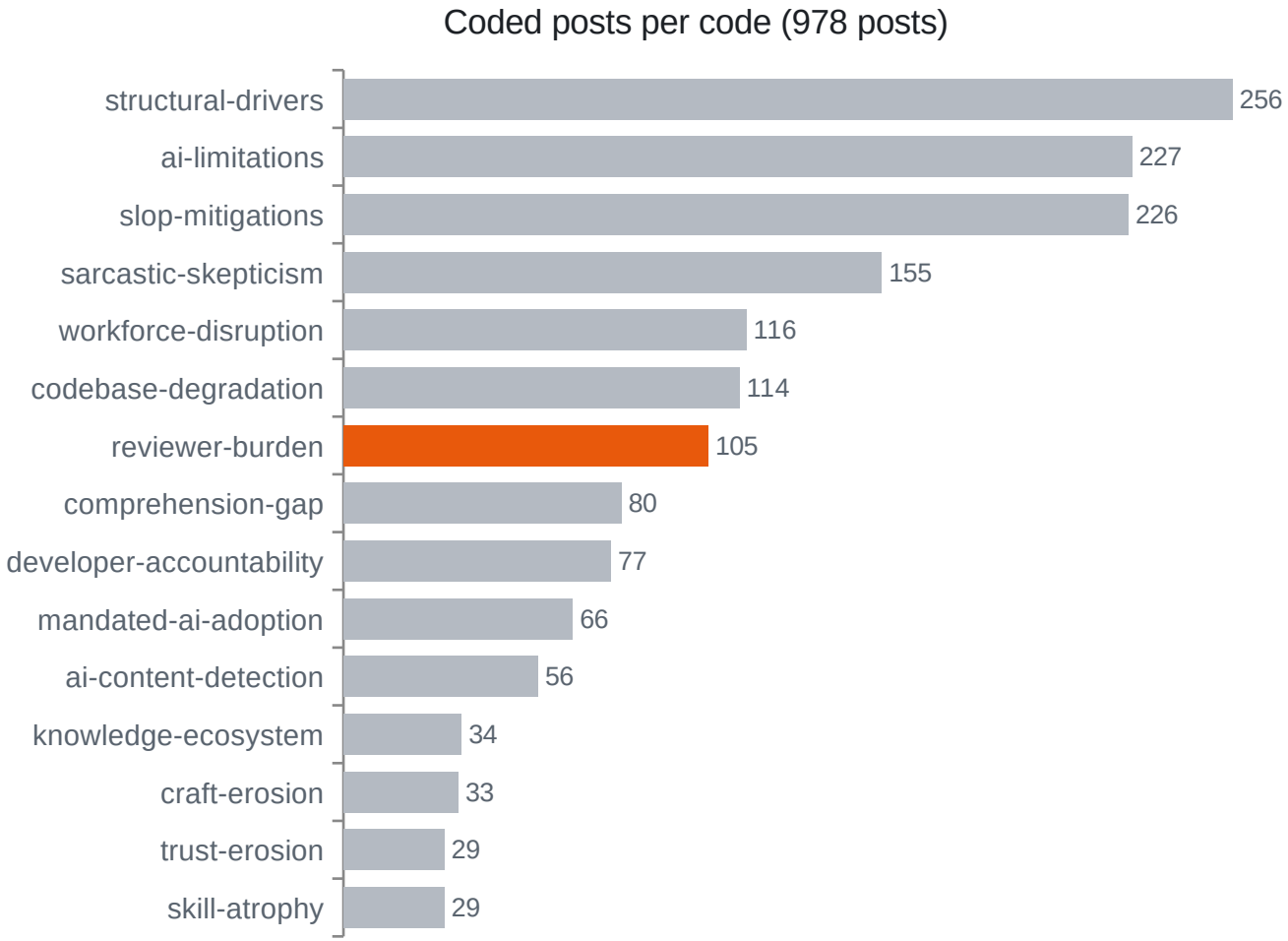
Forces and consequences

`structural-drivers · mandated-adoption · craft-erosion · workforce`

- Metrics that count output: contribution graphs, bug bounties.
- AI tools mandated by management.

“What does saving an hour or two writing code … give me if I have to spend that same hour or two deshittifying it?”

Most posts are about why, not review



10.7%

of coded posts mention reviewer burden. It ranks 7th of 15.

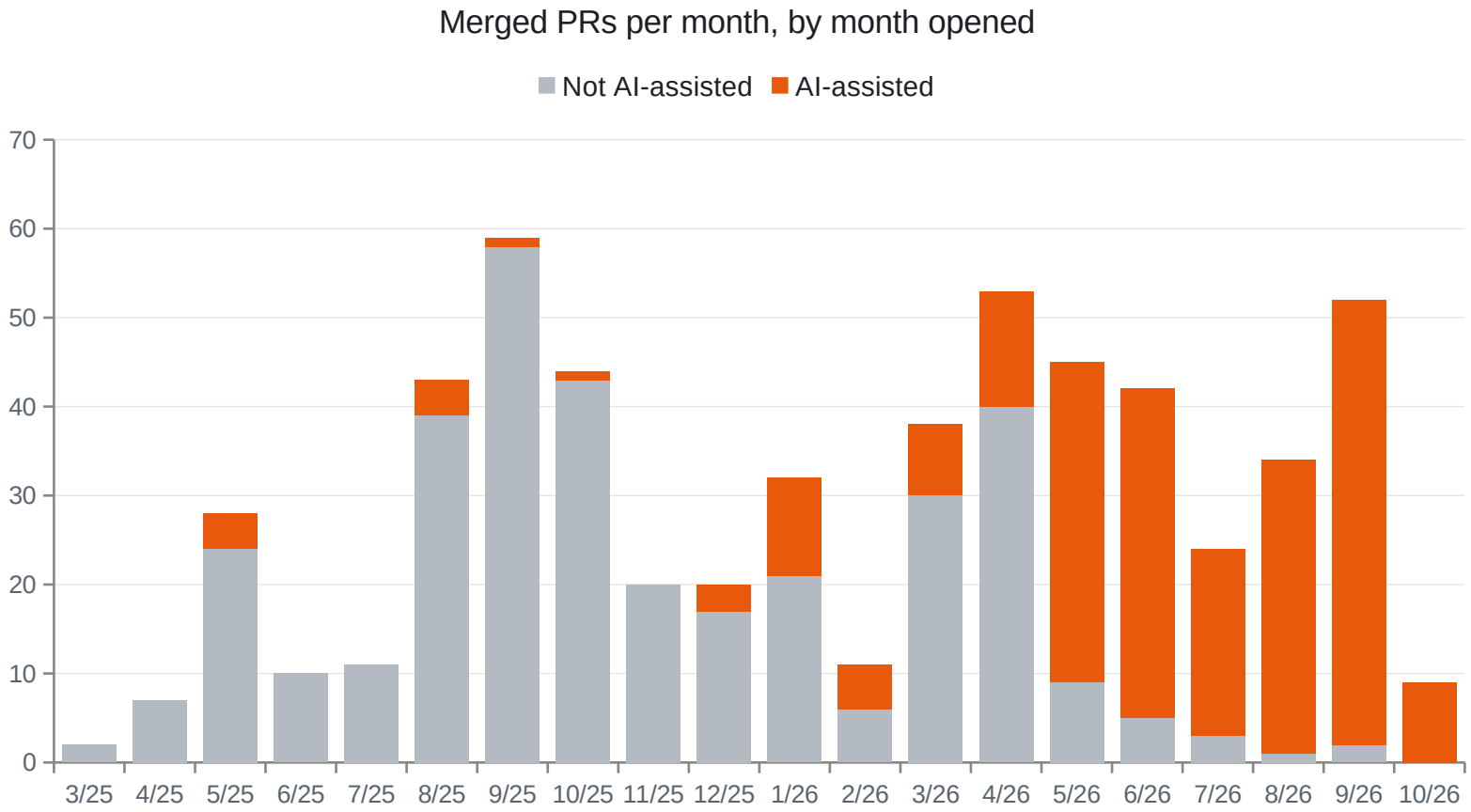
26%

mention structural drivers: metrics, mandates, speed targets.

The method maps the range of complaints, not how common they are. The authors say so themselves.

OUR DATA

Pawtograder went AI-first in May 2026



59 of 61

merged PRs opened in Sep–Oct 2026 were AI-assisted

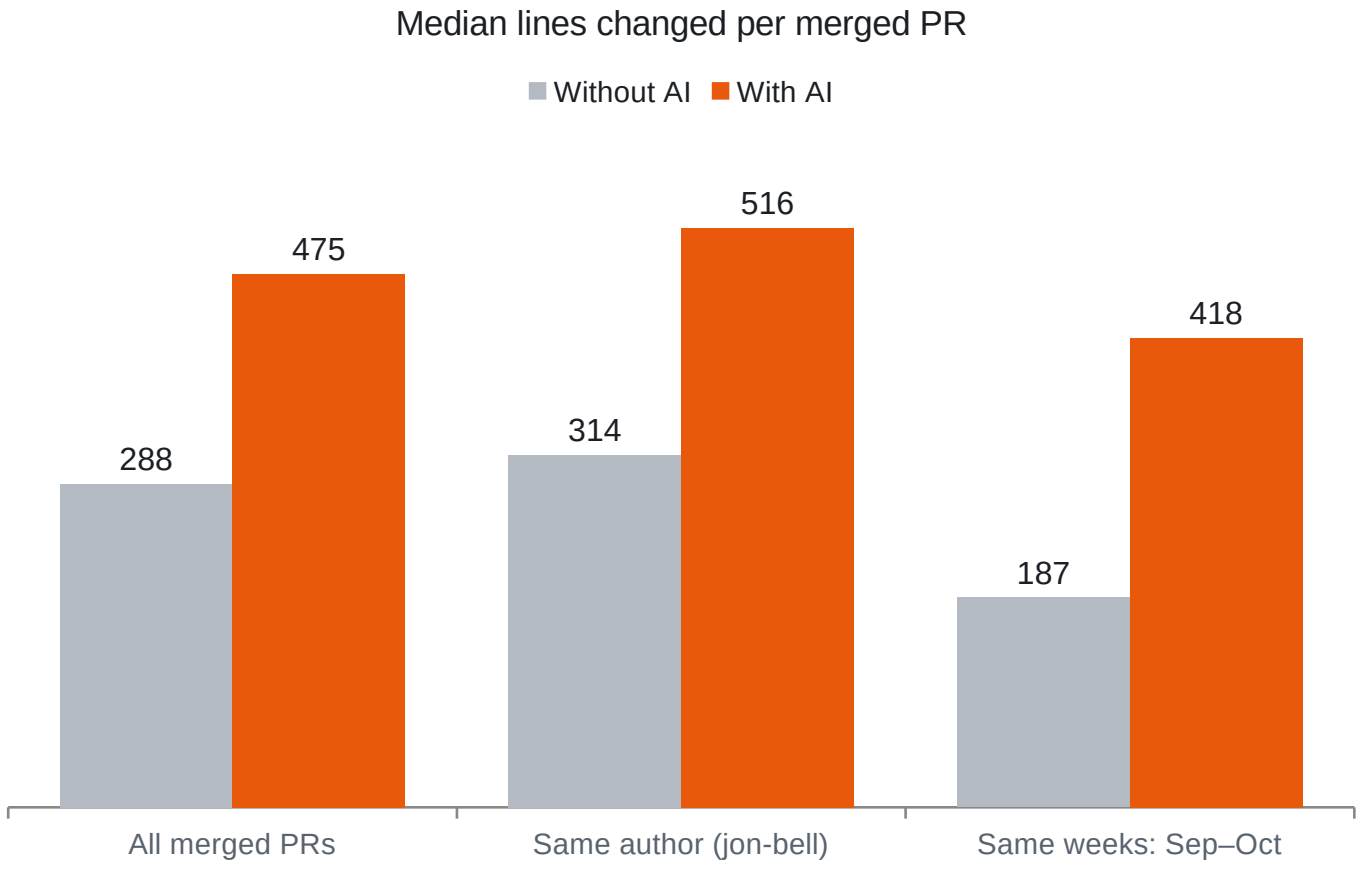
13 of 244

in all of 2025

584 merged PRs on pawtograder/platform, Mar 2025 – Oct 2026, pulled from the GitHub API on Oct 25, 2026. AI-assisted means disclosed: a Claude, Cursor, Copilot or Codex co-author trailer, “Generated with Claude Code” in the description, or a cursor/ or claude/ branch. Dependabot PRs excluded.

OUR DATA

AI-assisted PRs are bigger



2.2×

median PR size, Sep–Oct 2026 vs. Sep–Oct 2025

`[R05] · a mitigation`

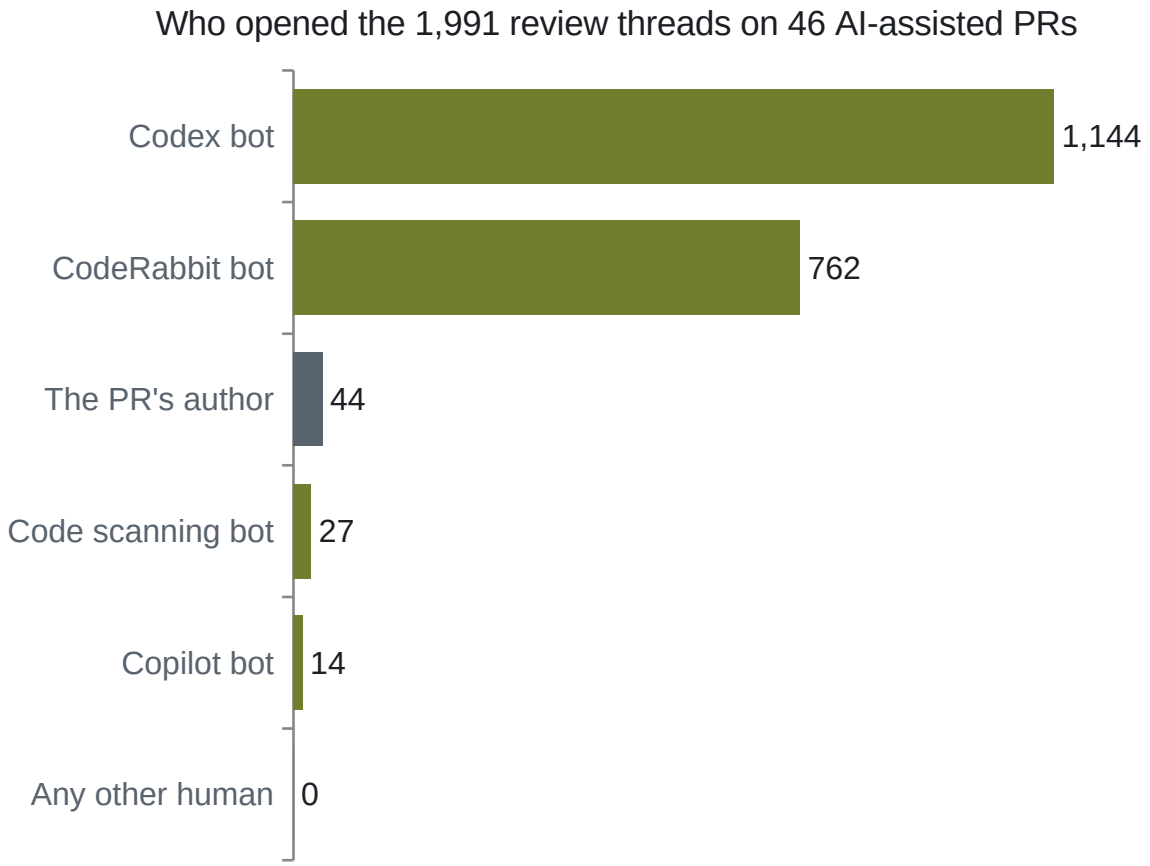
“less than 500 LOC per PR or they won't review it”

Our median AI-assisted PR is 475 lines. One in ten is over 5,383.

Lines changed = additions + deletions. jon-bell, the maintainer, opened 499 of the 586 merged PRs, so the middle pair compares one person with and without AI. In the Sep–Oct pair, “without” is 2025 (2 of 103 PRs AI-assisted) and “with” is 2026 (59 of 61).

OUR DATA

At Pawtograder, the reviewers are bots



5%

of AI-assisted merged PRs got any review from a human other than the author

4.1 vs 0.01

review threads per 1,000 lines: bots vs. other humans, on AI-assisted PRs

100%

of human replies to bot threads came from the PR's own author (739 threads answered, 38%)

Thread chart: the 46 AI-assisted PRs from 2026 with 15 or more bot threads. Other numbers: all 236 AI-assisted merged PRs.

OUR DATA

86,190 lines are waiting for review

24
open PRs

19
are AI-assisted

16
of those have no human review

9
open longer than 30 days

PR	What it does	Lines	Days open	Human reviews	AI
#1050	Bug reporter with session replay	40,279	25	**0**	yes
#814	Exam grading with OCR	10,360	143	3	yes
#778	Code coverage pipeline	4,964	157	**0**	yes
#722	Grading defaults and reminders	4,472	190	**0**	yes
#963	Keep a gradebook column's name	96	46	2	no

Open PRs on pawtograder/platform as of Oct 25, 2026. “Human reviews” counts reviews by anyone other than the author. #814 is the PR the code-review group showed last week.

OUR DATA

Two bot findings, checked by hand

PR #831 · CodeRabbit · Critical **REAL**

“Remove the orgName === "pawtograder" authorization bypass.”

Fixed before merge. But a human wrote that bypass in March 2025; this AI-assisted PR copied it into a new file. 4,571 lines, no human reviewer.

PR #936 · Codex · P1 **WRONG**

“…requires this nonstandard field while the mock fabricates it … masking the failure.”

Discord's docs say a bot install's token response does include a guild object. The thread is still unresolved. 13,577 lines, 64 bot threads, no human reviewer.

Slop can come from the reviewer too. Someone still has to check the checker.

SO WHAT

The paper's fixes, checked against us

What the paper's developers propose	Pawtograder today	Status
Cap PR size (under 500 lines)	Median AI-assisted PR is 475 lines; 1 in 10 is over 5,383	**Not in place**
Author self-reviews before peer review	Authors answer bot threads, but only 38% get a reply	**Partly**
A second reviewer from outside the team	5% of AI-assisted PRs get any non-author human review	**Not in place**
Tools that help verify, not just generate	CodeRabbit and Codex leave ~4 threads per 1,000 lines, and can be wrong (#936)	**In place**
Measure review effort and rework, not output	Not measured. We counted comments, not hours	**Not in place**

Proposals from the paper's Review Friction cluster ([R04], [R05]) and its implications section.

SKEPTICISM

Reasons to hold all of this loosely

THE PAPER

- Searching for “AI slop” finds people who are already annoyed
- It maps the range of complaints, not how common they are
- Reddit and Hacker News only, as of September 2025
- AI applied the codes; a human checked every one

OUR NUMBERS

- “AI-assisted” means disclosed. Earlier PRs may have used AI without saying so
- One author opened 499 of 586 merged PRs, so AI vs. not is mostly after vs. before May 2026
- We counted review comments, not reviewer hours
- Two bot findings checked by hand is a story, not an accuracy rate

DISCUSSION

AI saves a Pawtograder author two hours but adds an hour of review for another team. The author understands and checks the code. The other team already has PRs waiting. Is that a gain worth accepting, or should the author reduce the review burden, even if that uses up the time AI saved?

Take the two hours

The author checked it. Review is the reviewer's job.

Spend them on the reviewer

Smaller PRs, walkthroughs, a size cap.

Our twist: at Pawtograder the “other team” is mostly bots and one maintainer. Does that change your answer?